

5 MARKET OVERVIEW

5.1 INTRODUCTION

The microbiome sequencing services market is estimated to grow at a CAGR of 14.3% during the forecast period. Growth in this market is majorly driven by the increasing demand for metagenomic sequencing, the growing focus on human microbiome therapeutics development, and the high cost of advanced infrastructure for sequencing. However, a dearth of skilled personnel is expected to hinder the growth of this market during the forecast period.

5.2 MARKET DYNAMICS

FIGURE 18 MICROBIOME SEQUENCING SERVICES MARKET: DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

 DRIVERS	▪ Increasing demand for metagenomic sequencing ▪ Growing focus on human microbiome therapeutics development ▪ High cost of advanced infrastructure for sequencing
 RESTRAINTS	▪ Dearth of skilled personnel
 OPPORTUNITIES	▪ Expanding applications of microbiome sequencing in therapeutics development
 CHALLENGES	▪ Issues related to sample preparation/isolation of microbes for sequencing

Source: Secondary Literature, Interviews with Experts, and MarketsandMarkets Analysis

TABLE 4 MICROBIOME SEQUENCING SERVICES MARKET: IMPACT ANALYSIS OF DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES

FACTOR	IMPACT		
	2-3 YEARS	3-5 YEARS	5-10 YEARS
DRIVERS			
Increasing demand for metagenomic sequencing	High	High	High
Growing focus on human microbiome therapeutics development	Medium	High	Medium
High cost of advanced infrastructure for sequencing	Medium	Medium	Medium
RESTRAINTS			
Dearth of skilled personnel	High	Medium	Medium
OPPORTUNITIES			
Expanding applications of microbiome sequencing in therapeutics development	Medium	Medium	High
CHALLENGES			
Issues related to sample preparation/isolation of microbes for sequencing	Medium	Low	Low

5.2.1 DRIVERS

5.2.1.1 Increasing demand for metagenomic sequencing

Metagenomic sequencing is a target-independent approach that offers a comprehensive view of pathogenic agents and enables the detection of common and unexpected pathogens in samples. In recent years, there has been a significant increase in the demand for metagenomic sequencing in the field of microbiome research owing to its wide array of applications, such as disease diagnosis, drug discovery, and precision medicine. Recent developments in metagenomic sequencing technologies, such as improvements in sequencing accuracy, scalability, and data analysis, support the use of metagenomic sequencing. The development of long-read sequencing, single-molecule real-time sequencing, and nanopore sequencing has been particularly helpful in the metagenomic sequencing market. The demand for long-read sequencing-based metagenomics has increased in samples that are less complex, involve host-associated microbiomes, and comprise fewer microbial strains than microbiomes in marine environments, sediment, and soil. Such technologies are expected to drive the demand for microbiome sequencing services.

Over the years, the demand for precision medicine has increased. Precision medicine involves tailoring medical treatments and interventions to an individual's unique characteristics, including their microbiome. Metagenomic sequencing services are helpful in gathering essential information for personalized medicine, as well as identifying specific microbial signatures associated with disease risk and specific treatment responses. Metagenomic sequencing is also being used to study the role of microbiomes in cancer development, progression, and response to treatment. As per a study published in the Lancet in 2022, the gut microbiome of patients with metastatic melanoma can influence their response to immune checkpoint inhibitors. The study identified specific microbial species that were associated with better response rates.

and longer progression-free survival, suggesting that microbiome-targeted interventions could enhance the efficacy of cancer immunotherapies.

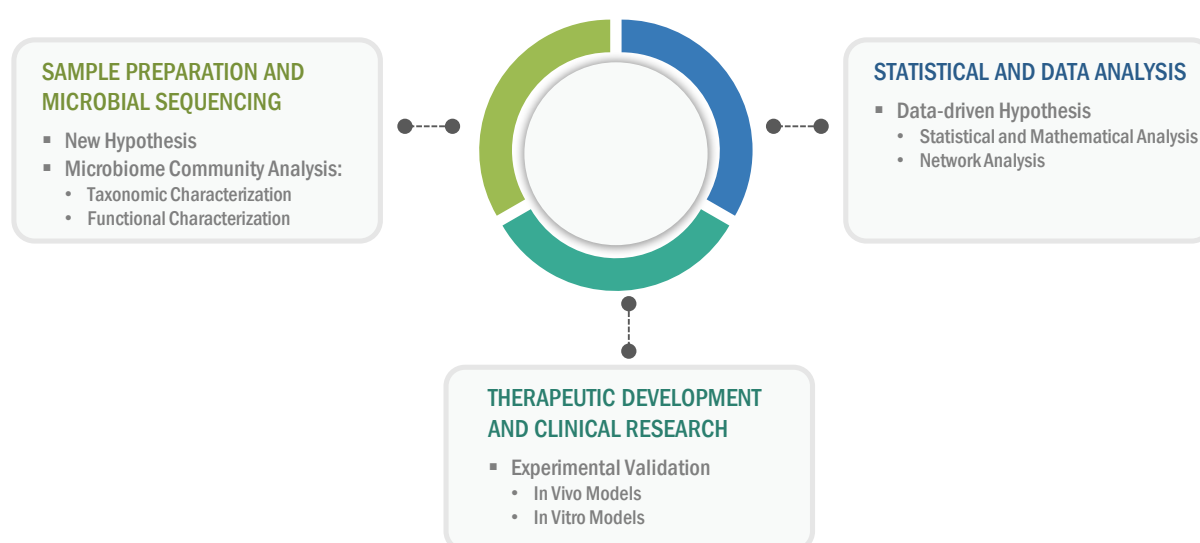
The demand for metagenomic sequencing has been driven by its applications in precision medicine, gut-brain axis research, and recent developments in sequencing technologies and data analysis methods. Metagenomic sequencing is proving to be a valuable tool for studying complex microbial communities and understanding their roles in various fields of research. The emergence of metagenomics has resolved the challenge of capturing the entire microbial community in different sample types by enabling the analysis of whole genomes without the need to culture the samples first. This is expected to boost the growth of the microbiome sequencing services market.

5.2.1.2 Growing focus on human microbiome therapeutics development

The increasing focus on human microbiome therapeutics development is driving therapeutics manufacturers to opt for sequencing, sample preparation, and data analysis services. This is further propelled by the need for a deeper understanding of the microbiome, including its composition, function, and interaction with the host, to develop microbiome-based therapeutics. Microbiome sequencing plays a critical role in providing this understanding by allowing for the identification and characterization of the microorganisms that make up the microbiome. Medium-sized and small-scale therapeutic manufacturers often prefer to outsource the initial stages of microbiome therapeutics development, such as screening, sample preparation, sequencing, and data analysis. This has offered significant momentum to large-scale CROs and service providers to develop their sequencing portfolio.

Human gastrointestinal microbiome research has the potential to offer pivotal clinical insights and facilitate the therapeutic development workflow. A research study conducted in June 2021, funded by the Australian National Health and Medical Research Council, highlighted a suitable avenue to translate microbiome research workflow, including the compositional and functional characterization of the microbiome, data-driven hypotheses generation, and the experimental validation of the hypotheses. By integrating different sequencing technologies (such as shotgun, metagenomic, and multi-omics) and data processing solutions, the ability to identify causative agents and design the appropriate microbiome therapeutics in disease treatment is expected to accelerate. The diagram below highlights the role of microbiome sequencing in such workflows.

FIGURE 19 ROLE OF MICROBIOME SEQUENCING IN THERAPEUTICS DEVELOPMENT



Source: Secondary Literature and MarketsandMarkets Analysis

Over the last few years, microbiome-based therapeutics development has witnessed significant advancements ranging from prebiotics and probiotics development to live biotherapeutics and microbiome mimetics. Advancements in bacterial culture methods from gastrointestinal (GI) samples, along with the expanding applications of metagenomic sequencing, have overcome several technical limitations in microbiome-based therapeutics development. However, a persisting challenge in this area is the development of appropriate methods to identify, refine, and test candidate therapies for microbiome medicines. Market players are progressing toward optimizing methods to identify candidate microbes, develop appropriate preclinical validation models, and progress to the personalized targeting of microbiome-based medicines through sequencing capabilities. This is expected to propel the market growth.

5.2.1.3 High cost of advanced infrastructure for sequencing

The field of microbiome sequencing requires sophisticated and expensive equipment, such as next-generation sequencing (NGS) platforms, bioinformatics tools, and data storage and analysis capabilities. Setting up and maintaining such infrastructure can be cost-prohibitive for many research and academic institutes, pharmaceutical and biotechnology companies, and other end users. Outsourcing microbiome sequencing to specialized service providers allows professionals to access advanced infrastructure and expertise without the need for substantial upfront investments. These service providers typically have state-of-the-art facilities and experienced personnel who are skilled in performing microbiome sequencing using the latest technologies and methodologies. Through outsourcing, professionals can leverage the capabilities of these specialized facilities without the burden of capital expenditures, operational costs, and maintenance expenses associated with in-house infrastructure.

The cost of NGS platforms is the highest in North America, followed by Europe. The cost in all regions is expected to increase by 2027 due to the high cost of raw materials, manufacturing, and labor. The cost of NGS platforms is low in the Middle East, Africa, and South America compared to Europe, North America, and the Asia Pacific due to the low cost of raw materials and the availability of a low-cost workforce (which reduces the overall manufacturing cost).

TABLE 5 AVERAGE SELLING PRICE OF NGS SYSTEMS, BY KEY PLAYER (2021)

COMPANY	INSTRUMENT	AVERAGE COST RANGE (USD)
Illumina	NovoSeq System	~850,000–900,000
Illumina	NextSeq System	~200,000–250,000
Illumina	MiSeq System	~80,000–85,000
Illumina	iSeq System	~18,000–20,000
Illumina	Mini Seq System	~40,000–45,000
PacBio	Sequel System	~350,000–370,000
PacBio	Sequel IIe System	~490,000–500,000

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.2.2 RESTRAINTS

5.2.2.1 Dearth of skilled personnel

The shortage of skilled personnel with expertise in sequencing and data analysis is a significant challenge in the field of microbiome sequencing. Microbiome research involves complex methodologies, including sample collection, DNA extraction, library preparation, sequencing, and data analysis. The proper execution of these steps requires specialized knowledge and technical expertise. However, the field of microbiome sequencing is relatively new and evolving rapidly, with a shortage of skilled personnel with the necessary expertise. Training and developing skilled personnel in microbiome sequencing can be time-consuming and resource-intensive, and there may be a lag in keeping up with the latest advancements in sequencing technologies and data analysis methods.

5.2.3 OPPORTUNITIES

5.2.3.1 Expanding applications of microbiome sequencing in therapeutics development

Microbiome sequencing is expanding its applications beyond research and is increasingly being utilized in therapeutics development, including the study of the skin microbiome and oral microbiome sequencing. These applications have the potential to advance the understanding of the role of microbiota in health and disease and inform the development of targeted interventions for improving skin health and oral health.

Skin microbiome sequencing has the potential to unravel the complex interactions between the skin microbiome and the host immune system to identify key microbial species or functional genes associated with specific skin conditions and elucidate mechanisms underlying the therapeutic effects of probiotics or other interventions. It can help inform the development of novel therapeutic strategies, including probiotics, prebiotics, and targeted antimicrobial treatments aimed at modulating the skin microbiome to improve skin health. Additionally, the speedy expansion of gut microbiome research activities has provided a foundation to explore the role of microbiota in other physiological systems, including the skin. Skin microbiome sequencing is targeted toward understanding the temporospatial distribution of cutaneous microbial communities. Research on cutaneous microbial communities is anticipated to provide crucial insights into microbiota in skin aging and pathology. Microbiome sequencing technologies are set to gain significant demand to overcome the limited information on correlative analysis, host-microbe interactions in the skin, and low taxonomic resolution. This is expected to open new avenues of business expansion for service providers operating in this market.

Oral microbiome sequencing can provide insights into the microbial etiology of oral health conditions, identify key pathogenic species or functional genes associated with oral diseases, and reveal potential therapeutic targets for intervention. This knowledge can be used to develop novel strategies for oral health management, including probiotics, antimicrobial treatments, and personalized oral care regimens tailored to an individual's oral microbiome.

5.2.4 CHALLENGES

5.2.4.1 Issues related to sample preparation/isolation of microbes for sequencing

Despite the success of using metagenomic sequence data to guide the isolation and cultivation of microbes, there are still challenges and limitations that need to be addressed for more widespread success. Anaerobic microbes may also pose technical challenges for some targeted isolation methods due to limitations with cell sorters. Determining suitable growth media and physicochemical conditions for target organisms based solely on genome sequences can be challenging, as natural environment compositions are often unknown. Contamination can also be a significant challenge in microbiome research, as it can introduce non-target microbial DNA from external sources during sample collection, processing, and sequencing. Contamination can arise from reagents, equipment, and environmental sources, leading to false-positive or false-negative results. Proper controls, including negative controls and rigorous contamination prevention measures, should be implemented to minimize contamination and ensure accurate and reliable sequencing results.

Recovering genomic data from target microbes for cultivation strategies can also be impacted by issues such as incomplete cell extraction, cell lysis, or DNA degradation, which may affect the representation of certain species in metagenome data. DNA extraction is a critical step in sample preparation for microbiome sequencing, and different DNA extraction methods can yield different results. Some DNA extraction methods may be biased towards certain microbial groups or may not effectively lyse certain tough-to-break cells, resulting in the incomplete representation of the microbial community in the sequencing data.

5.3 TECHNOLOGY ANALYSIS

Sequencing by Synthesis (SBS)

Sequencing by synthesis (SBS) is a widely used technology in microbiome sequencing services for analyzing microbial communities. SBS is a high-throughput DNA sequencing method that enables the rapid and cost-effective sequencing of millions of DNA fragments in parallel. It has revolutionized the field of microbiome research by allowing researchers to explore the diversity and composition of microbial populations in each sample with high resolution.

The basic principle of SBS involves the use of DNA polymerases, enzymes that can synthesize DNA strands, to sequentially add nucleotides to a growing DNA chain. As each nucleotide is added, it is labeled with a fluorescent dye that corresponds to the base being incorporated. The fluorescence signal is then detected and recorded, allowing the identification of the nucleotide that was added. This process is repeated multiple times, typically in cycles, to generate DNA sequences that can be assembled into a complete microbial community profile.

Sequencing by Ligation (SBL)

Sequencing by ligation (SBL) is a DNA sequencing technology that is used in microbiome sequencing services for analyzing microbial communities. SBL is a high-throughput DNA sequencing method that relies on the ligation of short DNA oligonucleotide adapters to target DNA fragments, followed by their sequential hybridization and ligation to complementary oligonucleotides, enabling the determination of the DNA sequence.

The basic principle of SBL involves the use of DNA adapters that are ligated to the DNA fragments to be sequenced. These adapters contain known sequences at their ends that serve as priming sites for the sequencing reaction. The DNA fragments, together with the adapters, are then subjected to multiple rounds of hybridization and ligation with a set of fluorescently labeled oligonucleotides that are complementary to the known adapter sequences. The ligation events are detected and recorded, allowing the determination of the DNA sequence.

Pyrosequencing

Pyrosequencing, also known as 454 sequencing, is a DNA sequencing technology that has been used in microbiome sequencing services for studying the composition and function of microbial communities. Pyrosequencing is a high-throughput NGS method that allows for the rapid and parallel sequencing of DNA fragments, providing deep insights into the genetic diversity of microbiomes.

The basic principle of pyrosequencing involves the detection of pyrophosphate (PPi) released during DNA synthesis. In this technology, DNA fragments are first amplified using PCR with primers that contain unique barcodes or indexes to enable sample multiplexing. The amplified DNA fragments are then sequenced in a massively parallel manner using a DNA polymerase enzyme and a series of nucleotide substrates that are added sequentially. As each nucleotide is added, the enzyme incorporates it into the growing DNA chain, and if the added nucleotide is complementary to the template DNA, pyrophosphate (PPi) is released as a byproduct. The released PPi is then converted into light, which is detected and quantified, allowing for the determination of the DNA sequence.

Sanger Sequencing

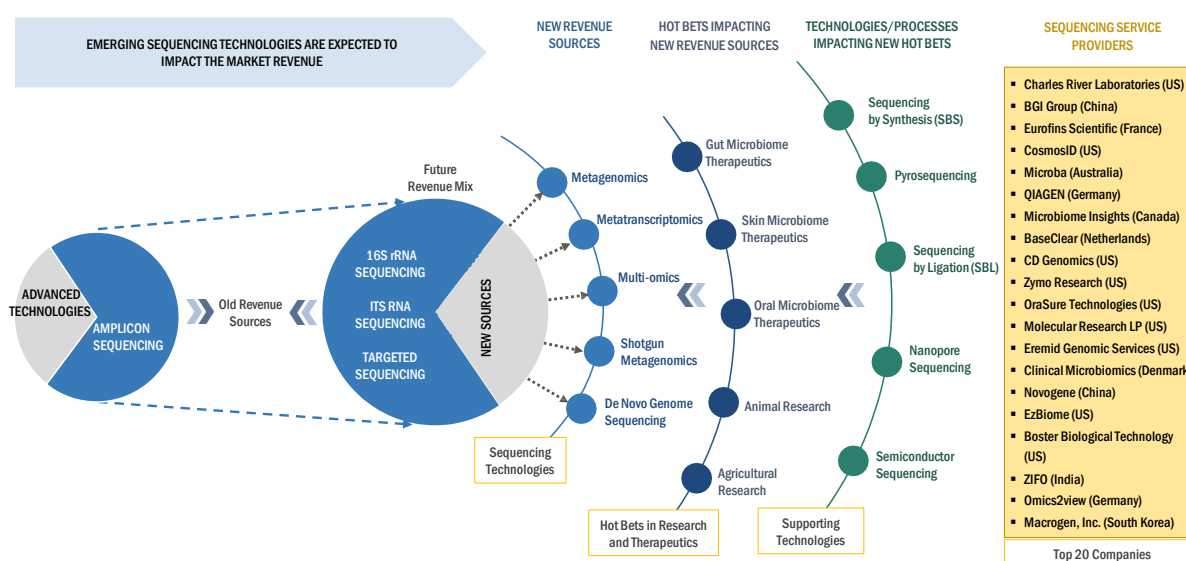
Sanger sequencing, also known as chain termination sequencing, is a traditional DNA sequencing technology used in microbiome sequencing services. While newer NGS technologies have gained popularity in recent years, Sanger sequencing still has its unique advantages and continues to be used in some microbiome research applications.

The basic principle of Sanger sequencing involves the enzymatic synthesis of DNA strands, where dideoxynucleotides (ddNTPs) are incorporated alongside regular deoxynucleotides (dNTPs) during DNA chain elongation. The incorporation of ddNTPs randomly terminates DNA synthesis, resulting in a set of DNA fragments of varying lengths that are complementary to the template DNA. The resulting DNA fragments are then separated by size using capillary electrophoresis, and the sequence is determined based on the order of the terminated ddNTPs.

5.4 TRENDS AND DISRUPTIONS IMPACTING CUSTOMERS' BUSINESSES

Robust research activities to explore the potential of the human microbiome present great potential in a number of treatments against obesity, inflammatory bowel disease, and diabetes, among other conditions. Information on microbiomes helps in the identification of the causal links between the microbiome and disease, further facilitating the development of efficient and targeted treatment strategies. Additionally, progress in state-of-the-art sequencing approaches, computational analyses, and experimental assays is supporting the growing demand for microbiome sequencing.

FIGURE 20 REVENUE SHIFT AND NEW REVENUE POCKETS FOR MICROBIOME SEQUENCING SERVICE PROVIDERS



Source: MarketsandMarkets analysis

The most commonly used sequencing technologies include 16S rRNA sequencing, ITS RNA sequencing, and whole-genome sequencing. Sequencing and experimental analyses of the microbiome continue to advance substantially. By coupling multi-omics technologies, statistical and computational analyses, and more advanced disease models, these approaches promisingly provide the opportunity to establish disease causality and subsequently inform therapeutic development.

5.5 VALUE CHAIN ANALYSIS

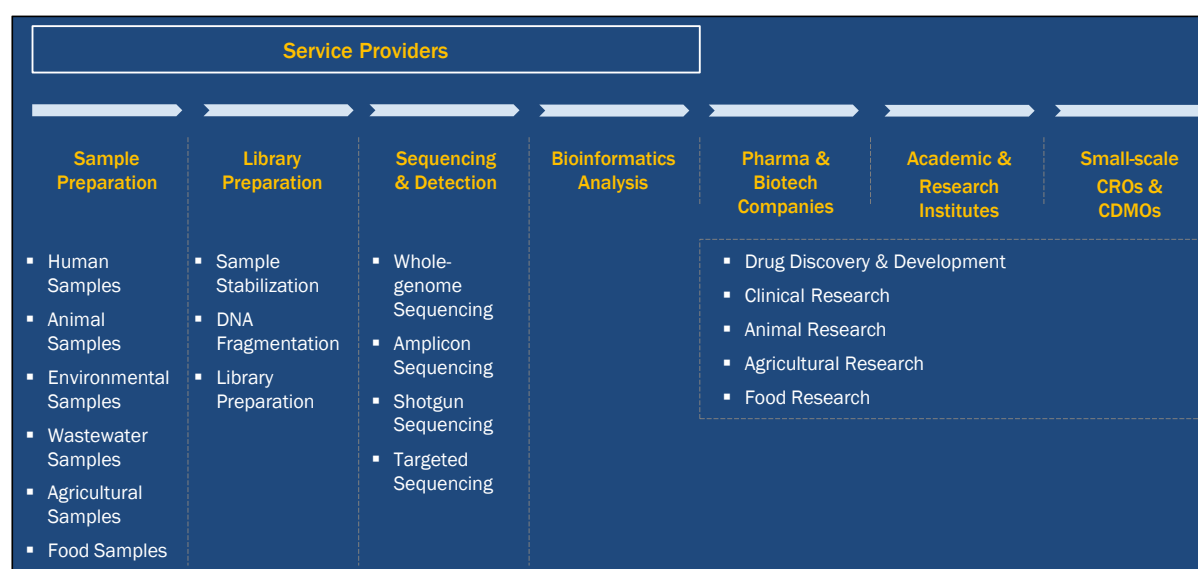
SAMPLE PREPARATION AND STUDY DESIGN SERVICES

Microbiome sequencing services start with outsourcing of study design, wherein the service provider helps in the implementation and validation of the hypothesis and the variables of interest in a sequencing project. Different variables in the aspect of study design include the number of samples required, apt sequencing technology to be used, and the requirement of follow-up studies. Service providers have expertise with respect to the collection, storage, and transport options optimized for this workflow.

SEQUENCING, LIBRARY PREPARATION, AND DATA ANALYSIS SERVICES

NGS instruments are predominantly used for the sequencing of DNA and RNA samples. Shotgun metagenomics sequencing is offered by a majority of sequencing service providers, even for deep-sequenced samples. DNA library preparation is verified for quality and yield. Genomic DNA is fragmented at specific sizes suitable for the purification/separation of desired proteins and nucleic acids. The selected fragments are then polished, A-tailed, and ligated with the full-length adapter. NGS technology is employed to sequence the sample, and the final stage involves bioinformatics analysis.

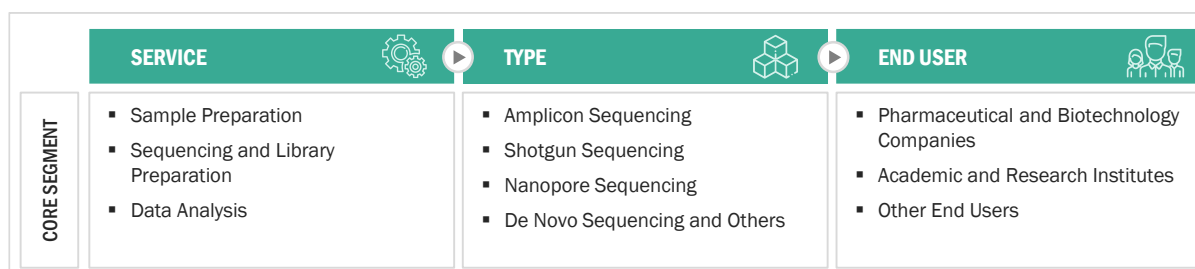
FIGURE 21 VALUE CHAIN ANALYSIS OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis

5.6 ECOSYSTEM MARKET MAP

FIGURE 22 ECOSYSTEM OF MICROBIOME SEQUENCING SERVICES MARKET



Source: MarketsandMarkets Analysis

5.7 PORTER'S FIVE FORCES ANALYSIS

TABLE 6 MICROBIOME SEQUENCING SERVICES MARKET: PORTER'S FIVE FORCES ANALYSIS

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	Low
Bargaining Power of Suppliers	Moderate
Bargaining Power of Buyers	Moderate
Threat of Substitutes	Low to moderate
Threat of New Entrants	Low to moderate

Source: Annual Reports, SEC Filings, Investor Presentations, Expert Interviews, and MarketsandMarkets Analysis

5.7.1 THREAT OF NEW ENTRANTS

The global microbiome sequencing services market is not a consolidated market. The well-established players operating in this market have narrow offerings, and a very small part of their portfolios are dedicated to microbiome sequencing. However, strong distribution channels and a global presence have made it easier for these players to maintain their market position. Moreover, existing product manufacturers have strong relationships with their customers (service providers) and are engaged in long-term contracts to supply the necessary equipment. As the industry demands high capital investments for R&D, testing, and manufacturing products, the entry barriers for new service providers in the market are estimated to be low to moderate.

5.7.2 THREAT OF SUBSTITUTES

The impact of this factor is low to moderate since there are fewer substitutes for established technologies in this market. The preference for traditional technologies for sequencing microbiomes may decrease due to the availability of advanced technology-oriented services.

5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers is moderate in this market. This market has witnessed significant technological innovations in developing affordable and user-friendly products. Thus, suppliers have better bargaining powers if their products are highly differentiated and endowed with breakthrough technologies. Products with a high degree of technological capabilities can command a premium price in the market. The high degree of technological capabilities, coupled with a high buyer-to-seller ratio, significantly increases the bargaining power of suppliers. Furthermore, compatibility issues between instruments and consumables (which are responsible for recurrent costs) increase the bargaining power of suppliers.

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers in the microbiome sequencing services market is moderate. Academic research institutes (one of the largest end users of sequencing products) are involved in bulk purchases of products and bulk sequencing of samples. This trend of bulk purchasing further increases the bargaining power of buyers in this market.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The microbiome sequencing services market is fragmented in nature, with the top 3 players accounting for a share of ~20%. A large number of local players and small-scale CROs account for the remaining market share. These players operate in niche market segments. Industry players are expected to cash in on the various growth opportunities and gain maximum share in the market. This has created low-entry barriers for new players, and the existing players compete to expand their shares.

5.8 REGULATORY ANALYSIS

5.8.1 NORTH AMERICA

5.8.1.1 US

NGS sequencing technologies are being used predominantly in microbiome and metagenome sequencing. The increase in use is further supported by increasing advancements in genomics. Several federal agencies regulate NGS products, including the Food and Drug Administration (FDA), the Centers for Medicare and Medicaid Services (CMS), and the Federal Trade Commission (FTC).

CMS Regulations:

The CMS regulates clinical laboratories, including laboratories conducting clinical genetic testing, through its Clinical Laboratory Improvement Amendments of 1988 (CLIA) program. The CLIA established a certification that laboratories must pass to conduct clinical testing legally. The objective of the CLIA is to determine clinical testing quality, including verification of the procedures used and the qualifications of the technicians processing the tests.

FDA Regulations:

Until recent years, the FDA has practiced enforcement discretion for laboratory-developed tests (LDTs), which implies that LDTs are used in the clinic without the FDA's assessment of their analytical and clinical validity. However, due to rapid advances in NGS, the growing need for clinical genetic testing, the growth of direct-to-consumer (DTC) genomic testing, and concerns that unregulated tests pose a public health threat, the FDA is modifying its approach.

The FDA has developed new guidance to regulate NGS genetic tests and verify their analytical and clinical validity. The agency has also drafted guidance proposing a new regulatory framework for LDTs. However, FDA guidance only represents the FDA's current thinking on a topic and is not legally binding for the FDA or the parties it regulates. However, adhering to FDA guidance is beneficial because it can streamline the regulatory process. The draft guidance is listed below. Since they are in draft form, they are not currently being implemented.

List of FDA Draft Guidance on Genetic Tests:

- Framework for Regulatory Oversight of Laboratory Developed Tests (2014)
- FDA Notification and Medical Device Reporting for Laboratory Developed Tests (2014)
- Discussion Paper on Laboratory Developed Tests (2017)
- Next-generation Sequencing (NGS) Draft Guidance

- Use of Standards in FDA Regulatory Oversight of NGS-based In Vitro Diagnostics Used for Diagnosing Germline Diseases

In addition, the US FDA classifies medical devices into various categories. It reviews NGS instruments as Class II medical devices. The NGS manufacturer must submit a premarket notification or 510(k) to prove substantial equivalence to a legally marketed device. If there is no substantially equivalent device in the market, the manufacturer must secure premarket approval (PMA) from the FDA.

The three main classifications of medical devices by the US FDA are as follows:

TABLE 7 US FDA: MEDICAL DEVICE CLASSIFICATION

CLASS I DEVICES	CLASS II DEVICES	CLASS III DEVICES
▪ Low-risk medical devices, which are subject to the least regulatory control ▪ Exempt from premarket notification [510 (k)]	▪ Devices that pose a greater risk than Class I devices; require reasonable assurance of safety and effectiveness ▪ Require premarket notification [510 (k)], with exceptions for certain devices	▪ Devices that are used for sustaining human life and are important for preventing the impairment of human health ▪ These are the highest-risk devices and, hence, are subject to the highest level of regulatory control ▪ Require submission of clinical trial data to the US FDA and require Premarket Approval (PMA) from the FDA

TABLE 8 US: MEDICAL DEVICE REGULATORY APPROVAL PROCESS

DEVICE CLASSIFICATION	WAITING PERIOD AFTER SUBMISSION UNTIL APPROVAL IS GRANTED	VALIDITY PERIOD FOR REGISTRATION OF THE DEVICE	WHETHER REGISTRATION RENEWAL SHOULD BE STARTED IN ADVANCE	COMPLEXITY OF THE REGISTRATION PROCESS	OVERALL COST OF GAINING REGULATORY APPROVAL
Class I Device	1 Month	No Expiry Date	NA	Simple	Low
Class II Device	6–9 Months	No Expiry Date	NA	Moderate	Medium
Class III Device	18–30 Months	No Expiry Date	NA	Complex	High

5.8.1.2 Canada

Health Canada is a key regulatory authority for medical devices in the country. The Canadian Medical Devices Regulation (CMDR) is the key directive that governs the approval of medical devices (including NGS systems) in the country. In Canada, medical device manufacturers must first register and obtain a proper license from Health Canada to market their IVD devices. Health Canada has established a four-tier, risk-based classification system (Class I, II, III, and IV) for IVD products.

5.8.2 EUROPE

From May 2020, Europe has implemented new Medical Device Regulation (EU-MDR) guidelines to increase the safety of the medical devices used in the region. These guidelines require manufacturers to make significant data reporting and quality assurance changes. While new medical devices would be required to adhere to the new guidelines, existing medical devices would have to be recertified (in some cases, they would also have to be reclassified) by manufacturers.

In Europe, medical devices have been classified into four categories based on their function, risk to patients, and the manufacturer's intended use.

- Class I: Provide non-sterile or do not have a measuring function (low risk) and provide sterile and/or have a measuring function (low/medium risk); the MDR has added reusable surgical instruments to this group
- Class IIa (medium risk)
- Class IIb (medium/high risk)
- Class III (high risk)

However, with changes in MDR regulations, some of these medical devices have been reclassified. The new MDR regulations have 22 rules for classification, and the scope of medical devices has increased, including additional products.

Three Medical Device Directives mainly regulate the European market—the Medical Device Directive (93/42/EEC), the Active Implantable Medical Device Directive (90/385/EEC), and the In Vitro Diagnostic Medical Device Directive (98/79/EC). The European Medical Device Directive's Conformité Européenne (CE mark) approval is required for the distribution and marketing of any medical devices (including NGS systems) in the European (EU) region. This certification verifies that a device meets all regulatory requirements of the Medical Devices Directive (MDD). The manufacturer must be certified by the notified body to Annex II, V, or VI of the Medical Device Directives and comply with the requirements of those directives. A certified body conducts a technical review, and if all requirements are met, a CE clearance letter is issued, permitting the sale of a particular medical device in Europe. Also, manufacturers of medical devices must implement quality systems compliant with ISO 13485:2003.

5.8.3 ASIA PACIFIC

5.8.3.1 China

The National Medical Products Administration (NMPA), formerly the CFDA, is responsible for supervising and administering medical devices in the People's Republic of China (PRC). The NMPA classifies medical devices into Class I, Class II, and Class III based on the safety and effectiveness of the device.

TABLE 9 CHINA: CLASSIFICATION OF MEDICAL DEVICES

DEVICE CLASSIFICATION	POST-SUBMISSION APPROVAL (MONTHS)	LEVEL OF RISK	COMPLEXITY OF REGISTRATION	OVERALL COST OF GAINING REGULATORY APPROVAL	DEFINITION
Class I	<1	Low	Low	Moderate	Devices for which safety and effectiveness can be ensured through routine administration
Class II	16–24	High	High	Moderate-to-high	Devices for which further control is required to ensure safety and effectiveness
Class III	16–24	High	High	High	Devices that are implanted into the human body or that are used for supporting or sustaining life. They pose a significant risk to the human body and, hence, must be strictly controlled for their safety and effectiveness

NGS systems are classified as Class II devices as per NMPA rules. In China, IVD products (including NGS systems) and their components that are recognized internationally are classified as medical devices (MD), in vitro diagnostic reagents (including NGS reagents), and drugs. For an IVD system that includes an instrument and appropriate reagents, such as NGS, two product registration certificates must be acquired for market approval—medical device registration for instruments and IVD reagent registration for suited reagents.

5.8.3.2 Japan

The Ministry of Health, Labour and Welfare (MHLW) and the Pharmaceuticals and Medical Devices Agency (PMDA) govern the regulatory approval pathway to assess the safety and effectiveness of medical products in the country. Medical device manufacturers must abide by the PMDA. Obtaining PMDA approval for a new product is time-consuming (it can take approximately five years) and cumbersome. The compilation of foreign medical device manufacturers with PMDA is challenging due to the multilayered regulatory and language barriers. Hence, foreign manufacturers (with no local presence in Japan) need to appoint a Marketing Authorization Holder (MAH) to manage their device registration process and liaise with Japan's Pharmaceuticals and Medical Devices Agency.

5.8.3.3 India

According to the Central Drugs Standard Control Organisation (CDSCO), diagnostic kits are classified as notified diagnostic assays. Manufacturers must register these tests with the regulatory authorities and secure an import/manufacturing license. The remaining molecular diagnostic assays are termed non-notified diagnostic assays and need only an import license to be sold in India. The Drugs and Cosmetics Act of 1940 and the Drugs and Cosmetics Rules of 1945 are the major laws that regulate pharmaceuticals, medical devices, and diagnostics in India. The Ministry of Health & Family Welfare, through the Drug Controller General of India, is responsible for the improvement of public health by monitoring and regulating the quality, safety, and efficacy of drugs and devices through examination and investigation.

5.8.4 LATIN AMERICA

In some Latin American countries, such as Mexico and Argentina, medical device regulations are evolving, while in others, the regulations are static and, to some extent, opaque. Most Latin American countries consider premarket controls the primary feature for medical device regulation (including NGS). Medical device regulations in this region are driven by trade agreements, primarily Mercado Común del Sur (MERCOSUR), which seeks to create a common market like that in the EU. Medical goods and services can move freely among member countries. To register products in a Latin American country, a company must have an office in that country or have a local distributor (registered with the Ministry of Health) to register products on its behalf. For all Latin American registrations, a Free Sale Certificate (FSC) or a Certificate to Foreign Government (CFG) is required to confirm that a product is approved in the country of origin and can be exported without restriction.

5.8.5 MIDDLE EAST

Each member country of the Middle East has its regulatory system for medical devices in different maturity stages. Saudi Arabia has the most established regulatory framework for medical devices among Middle Eastern countries, followed by the UAE and Qatar. The regulatory systems of the UAE and Qatar are still in the early stages of development. The ministries of health in each of these countries are responsible for regulating medical devices. Most countries need overseas device manufacturers to appoint a local (or authorized) representative to facilitate the registration/regulation (in addition to distribution and sale) of their products in each country.

A marketing authorization from the Saudi Food and Drug Authority (SFDA) under Interim Regulation (1-8-1429) is required to sell medical devices in Saudi Arabia. Medical device manufacturers must obtain prior market authorization in reference markets such as Australia, Canada, the European Union, Japan, or the US. The device classification in respective reference markets will determine the classification of that device in Saudi Arabia. An authorized representative primarily manages the device registration in Saudi Arabia. The representative must be licensed with the SFDA.

5.8.6 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS

TABLE 10 REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS: GENOMICS, MICROBIOME RESEARCH, AND METAGENOMICS

REGION	COUNTRY	REGULATORY BODIES, GOVERNMENT AGENCIES, AND OTHER ORGANIZATIONS
North America	US	- National Institute of Standards and Technology (NIST) - National Human Genome Research Institute (NHGRI) - Joint Genome Institute (JGI)
	Canada	- The Genomics Research and Development Initiative (GRDI) - Genome Canada
Europe	Germany	- DZIF German Centre for Infection Research - HELMHOLTZ-ZENTRUM FÜR INFektionsFORSCHUNG GMBH
	UK	- UK Research and Innovation (UKRI) - Microbiology Society
	France	- Paris Center for Microbiome Medicine (PaCeMM) FHU - Microbiome Foundation France
	Italy	- Microbiome Research Hub Italy - Council for Agricultural Research and Economics (CREA)
	Spain	- The Centre for Genomic Regulation (CRG) - National Centre for Genomic Analysis (CNAG-CRG) - The Higher Council for Scientific Research - The Institute for Research in Biomedicine (IRB)
Asia	Japan	- Human MetaGenome Consortium Japan (HMGJ) - Japan Agency for Medical Research and Development
	China	- Agricultural Genomics Institute at Shenzhen (AGIS) - National Genomics Data Centre China
	India	- Department of Biotechnology (GENOME India and Microbiome) - International Life Sciences Institute India (ILSI)
	South Korea	- National Research Foundation of Korea (NRF) - Korea Biotechnology Industry Organization (KoreaBIO)
	Australia	- National Research Infrastructure for Australia (Australian Microbiome Initiative) - Microbiome Research Centre (MRC)
Latin America	Brazil	- Brazilian Society of Genetics (Sociedade Brasileira de Genética in Portuguese) - The São Paulo Research Foundation (Fundação de Amparo à Pesquisa do Estado de São Paulo in Portuguese)

Middle East & Africa	Saudi Arabia	▪ The Saudi Society of Medical Genetics (SSMG) ▪ King Faisal Specialist Hospital and Research Centre's (KFSH&RC) Genetics department ▪ Medical Genetics Programme ▪ Center of Innovation in Personalized Medicine (CIPM)
	South Africa	▪ South African Medical Research Council (SAMRC) ▪ South African Society for Human Genetics ▪ Cancer Association of South Africa (CANSa) ▪ Hyrax Biosciences

Source: Government Websites and MarketsandMarkets Analysis

5.9 PRICING ANALYSIS

The pricing of microbiome sequencing services can vary depending on several factors, including the specific sequencing technology used, the depth of sequencing coverage, the number of samples, the data analysis and interpretation required, and the infrastructure capacity of the service provider.

TABLE 11 AVERAGE SELLING PRICE, BY SERVICE AND TYPE

SERVICE PROVIDER	REGION/COUNTRY	TYPE	SERVICE	SAMPLE SIZE/ CUSTOMER TYPE	PRICE RANGE
Dalhousie University Integrated Microbiome Resource (IMR)	North America	Amplicon Sequencing (16S/18S/ITS Short Amplicons)	Library Preparation + Sequencing	50 to 200 Samples	USD 20–30
			Sample Preparation (DNA Extraction)	50 to 200 Samples	USD 15–20
			Standard Bioinformatics Pipeline	50 to 200 Samples	USD 500–1,000
		Metagenome Sequencing	Library Prep. + Sequencing	Academic Customers	USD 100–150
			DNA Extraction	Academic Customers	USD 15–20
			Standard Bioinformatics Pipeline	Academic Customers	USD 1,000–1,200
Genotypic Technology	Asia Pacific	Metagenomic Analysis	Library Preparation + Sequencing	Per Sample	USD 20–25

		De Novo Transcriptome Sequencing	Library Preparation + Sequencing	Per Sample	USD 500–600
		De Novo Wholeenome Sequencing	Library Preparation + Sequencing	Per Sample	USD 2,500–3,000
MR DNA	North America	Shotgun Metagenomic Sequencing	Library Preparation + Sequencing	Per Sample	USD 50–60
			Sample Preparation (DNA/RNA Extraction)	Per Sample	USD 20–70
CD Genomics	North America	Metagenomic Analysis	Library Preparation + Sequencing + Data Analysis	Per Sample	USD 100–120
UConn CORE MARS	North America	Microbiome Analysis	Data Analysis	Per Sample	USD 100–150
Dreamgenics	Europe	Metagenomic Analysis	Data Analysis	Per Sample	USD 30–40

Note: This is an indicative list of pricing specific to the particular service provider operating in a region.

Source: Company Websites and MarketsandMarkets Analysis

5.9.1 AVERAGE SELLING PRICE TREND ANALYSIS

In recent years, the cost of microbiome sequencing has generally decreased due to the rapid advancements in NGS technologies, which have made high-throughput sequencing more affordable and accessible. The decreasing cost of DNA sequencing has enabled researchers to generate larger and more comprehensive microbiome datasets, leading to a greater understanding of microbial communities and their roles in various health and disease conditions. Pricing also varies geographically, with different regions or countries having different costs associated with microbiome sequencing services due to varying market dynamics and operational costs. The average cost of microbiome sequencing services is the highest in the US, followed by Europe and the Asia Pacific.

5.10 KEY CONFERENCES AND EVENTS IN 2023–2024

TABLE 12 MICROBIOME SEQUENCING/METAGENOMIC SEQUENCING/NGS CONFERENCES, 2023–2024

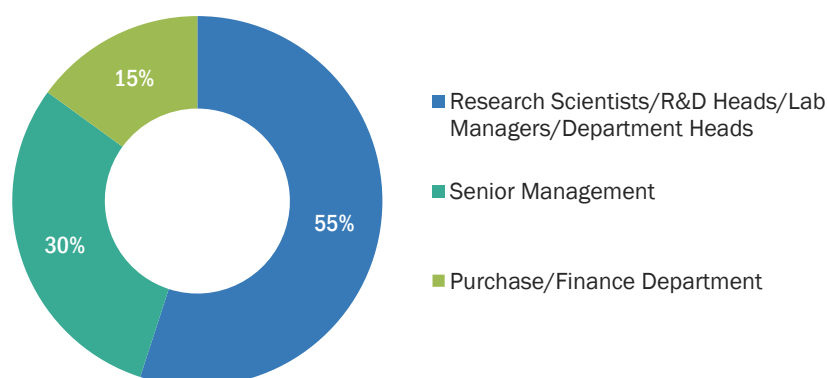
YEAR/QUARTER	NORTH AMERICA	EUROPE	ASIA PACIFIC	REST OF THE WORLD
Q1 2023	(February) International Conference on Next Generation Sequencing and Metagenomics (March) Annual Next Generation Sequencing US Congress	(January) Human Genomics and Genomic Medicine (February) International Conference on Next Generation Sequencing in Biology (March) Genomics and Molecular Biology (March) Conference on Revolutionizing Next Generation Sequencing (April) International Conference on Next Generation Sequencing Approaches in Biological Engineering	(January) International Conference on Next Generation Sequencing Approaches in Biology (February) International Conference on Next Generation Sequencing and Metagenomics	(February) International Conference on Next Generation Sequencing in Biology (April) International Conference on Next Generation Sequencing Technologies ICNGST
Q2 2023	-	-	(May) International Conference on Next Generation Sequencing in Biological Engineering	-
Q3 2023	(June) Microbiome Movement Drug Development Summit (July) International Conference on Next Generation Sequencing and Diagnostics	(June) International Conference on Microbiome Research (ICMR) (July) International Conference on Next Generation Sequencing Technology (July) International Conference on Next Generation Sequencing and Metagenomics	-	(July) International Conference on Next Generation Sequencing and Clinical Diagnostics

		(October)		
		International		
		Conference on		
		Bioinformatic		
Q4 2023	(December)	Approaches for	-	-
	Genomics &	Human Microbiome		
	Pharmacogenomics	(ICBAHM)		
		(November) NextGen		
		Omics UK		
		(March)		
		International		
		Conference on		
		Microbiome Analysis		
Q1 2024	-	(ICMA)	-	-
		(February)		
		International		
		Conference on Next		
		Generation		
		Sequencing and		
		Metagenomics		
Q2 2024	-	-	-	-
		(July) International		
		Conference on Next		
Q3 2024	-	Generation	-	-
		Sequencing		
		(November)		
		International		
Q4 2024	-	Conference on Next	-	-
		Generation		
		Sequencing		

Source: Org. Websites, Expert Interviews, and MarketsandMarkets Analysis

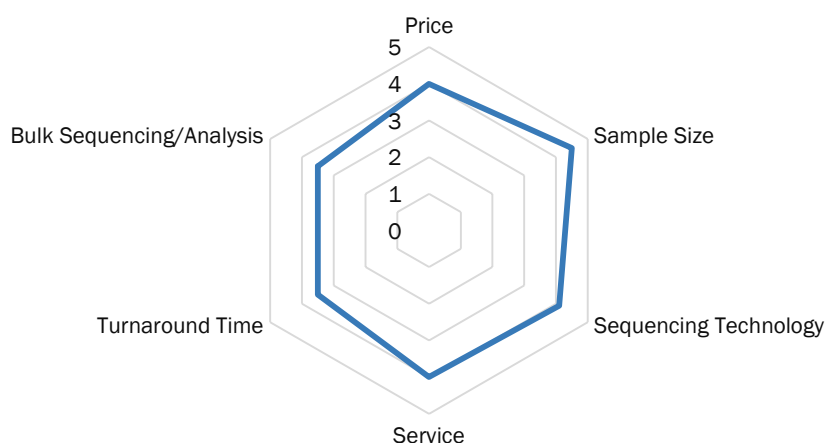
5.11 KEY STAKEHOLDERS AND BUYING CRITERIA

FIGURE 23 INFLUENCE OF STAKEHOLDERS ON BUYING PROCESS OF SEQUENCING SERVICES



Source: Expert Interviews and MarketsandMarkets Analysis

FIGURE 24 CRITERIA FOR OUTSOURCING SEQUENCING SERVICES



Note 1: Service refers to the preference for end-to-end services (sample preparation, sequencing, and data analysis) among end users.

Note 2: Bulk sequencing refers to the preference for discounted offers on bulk sequencing (sample size of more than 100) among end users.

Source: Expert Interviews and MarketsandMarkets Analysis